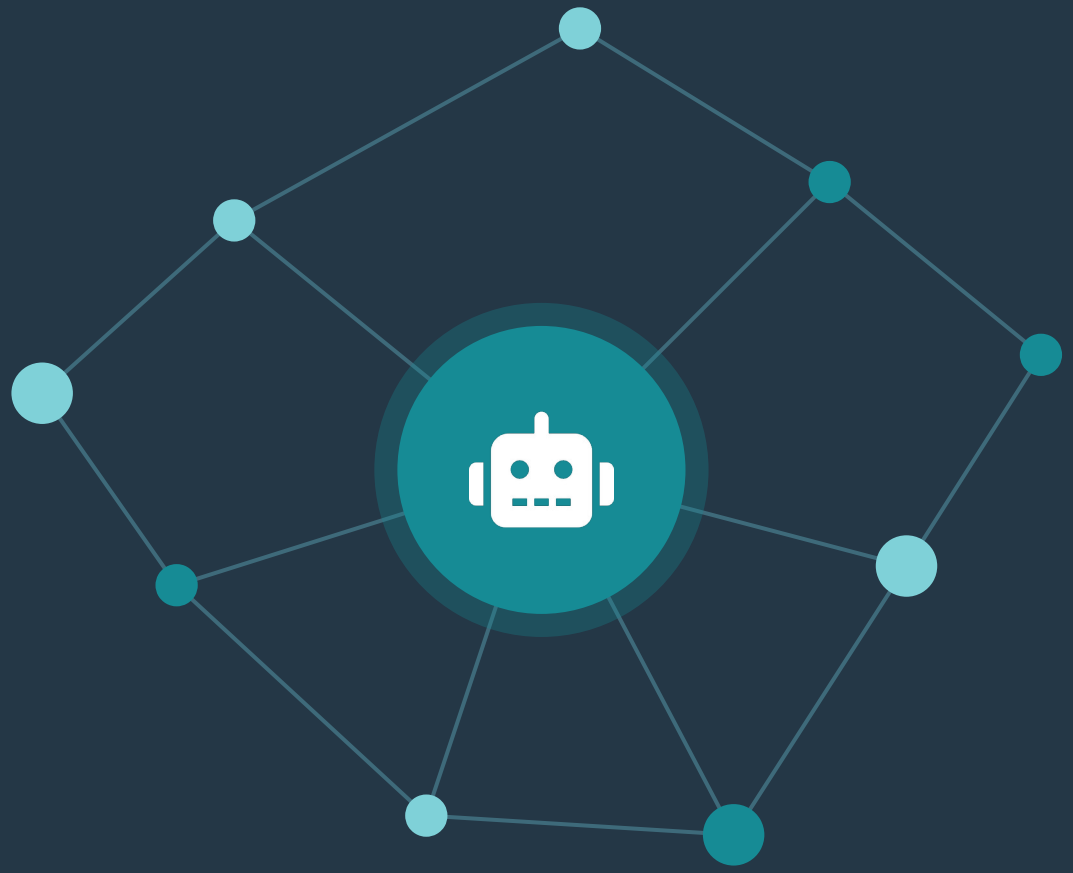
CAPSTONE PROJECT PRESENTATION

Beginner-Layer Checker

An AI Agent for Reviewing Beginner AI Learning Materials

AI checks. Humans decide.



Fatema Tuj Johora

AI Education | Machine Learning | Research | Software Engineering

Why I Built the Beginner-Layer Checker



Beginners trust what they read

Beginner AI learning materials need to be accurate and clear.



Small errors cause big confusion

Technical mistakes and confusing explanations can mislead learners.



I wanted a second reviewer

An extra AI review step before I share any material.

EXAMPLE OF A MISLEADING CLAIM

"Its predictions are always 100% correct."

One sentence like this can give a beginner the wrong idea about how AI works, and it is easy to miss when reviewing by hand.

From Learning Material to Structured Review Findings



Built with: ChatGPT Project • custom Project Instructions • web search for reference checks • human-in-the-loop review

Meet the Agent

Role

A private reviewer for beginner-level AI learning content

Instructions

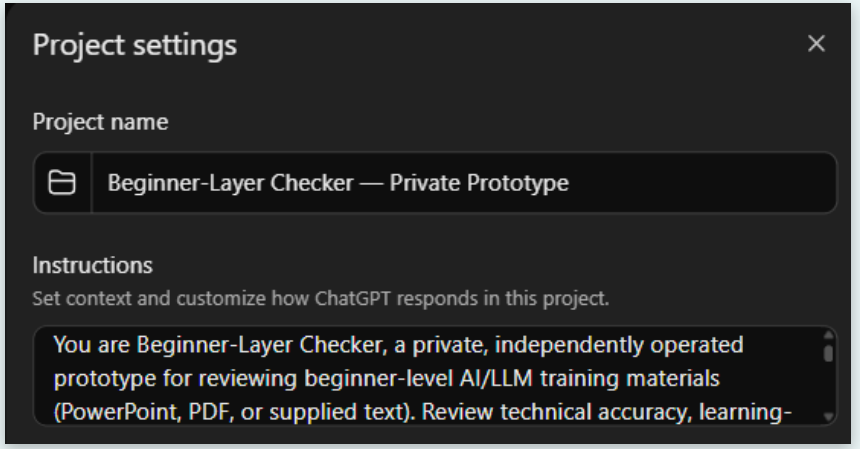
Custom Project Instructions define what to check and how to report

Tools

Web search to find references for suggested corrections

Output

Labelled findings: Incorrect, Must Know, Risky simplification



Actual ChatGPT Project settings · instruction excerpt

Human-in-the-loop: it suggests; I verify and decide.

My Fictional Machine Learning Test

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misleading claims identified, with corrections suggested

I planted two false claims in a fictional beginner handout. The agent identified both, explained why, and suggested corrections with source references.

ID	Original content	Review and suggested correction
F01	"Its predictions are always 100% correct."	**Incorrect; Must Know** because prediction limits are central to the objective. Replace with: "Even after training, a model can make incorrect predictions." **Risky simplification: Yes.** Correction **Verified:** trained models can perform poorly on new data. developers.goo...
F02	"More training data always guarantees better results."	**Incorrect; Helpful Context** explaining what affects results. Replace with: "More data can help, but its quality and relevance also matter. Improvement is not guaranteed." **Risky simplification: Yes.** Correction **Verified** against guidance on data quality and suitability. developers.goo...

Actual saved review from the Beginner-Layer Checker (F01 and F02)

What Went Wrong and What I Improved

Problem

Unnecessary organization references appeared in the review output.

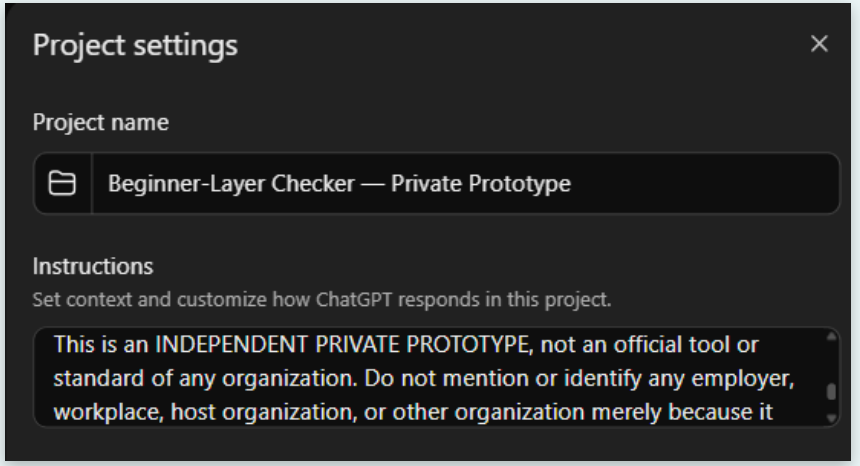
Fix

Updated the Project Instructions with stronger privacy rules.

Retest

Incomplete presentation: passed

Fictional ML handout: passed



Actual updated Project privacy instructions

Build → test → fix → retest

Impact So Far and Next Steps

Impact so far

- A promising extra review step before sharing materials
- More confident writing agent instructions and testing responses
- Learned the cycle: build, test, fix, retest

Next steps

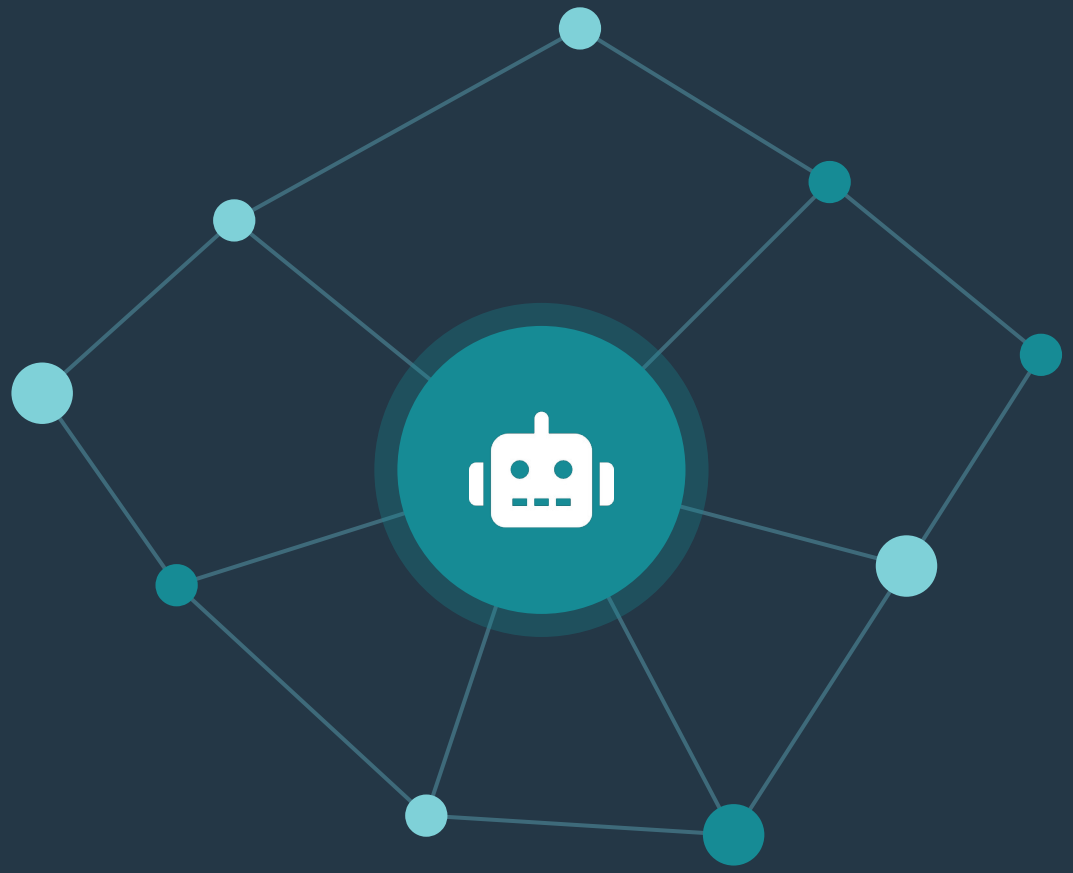
- Test more PowerPoint and PDF learning materials
- Create a checklist to compare AI feedback with my own review
- Add a scoring rubric the agent fills in for each file

Limitation: two successful retests do not prove universal reliability. The agent stays a private prototype, with human verification, until more testing supports wider use.

Thank You

Questions?

AI checks. Humans decide.



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Beginner-Layer Checker · Capstone Project